

Open Shortest Path First (OSPF)

Comprehensive Technical Study Guide & Verification Summary

1. Overview & Key Characteristics

OSPF is an Interior Gateway Protocol (IGP) based on Link-State technology. It uses Dijkstra's Shortest Path First (SPF) algorithm to calculate loop-free shortest paths to all destinations.

- **Protocol Type:** Link-State IGP (Open standard, RFC 2328).
- **Algorithm:** Dijkstra SPF algorithm.
- **Metric:** Cost, calculated as:

$$\text{Cost} = \frac{\text{Reference Bandwidth}}{\text{Interface Bandwidth}}$$

Default reference bandwidth is 10^8 bps (100 Mbps). Interfaces ≥ 100 Mbps have a default cost of 1 unless adjusted using `auto-cost reference-bandwidth`.

- **Administrative Distance (AD):** 110.
- **Classless:** Carries subnet masks in updates; fully supports VLSM and CIDR.
- **Hop Count:** Unlimited (avoids RIP's 15-hop limit).
- **Load Balancing:** Equal-Cost Multi-Path (ECMP) only. Default 4 paths, configurable up to 32.
- **Multicast Addresses:**
 - 224.0.0.5: All OSPF Routers (Hello packets, DR flooding).
 - 224.0.0.6: All Designated Routers (DROTHER to DR/BDR communication).
- **Transport:** Encapsulated directly into IP (IP Protocol 89). No TCP/UDP overhead.
- **Convergence:** Rapid, event-driven incremental updates using Link-State Advertisements (LSAs).

2. Hierarchical Area Design

OSPF segments Autonomous Systems into logical areas to restrict database size, limit SPF calculation overhead, and isolate flapping links.

- **Area 0 (Backbone Area):** Central transit area. All non-backbone areas must connect directly to Area 0.
- **Area Border Router (ABR):** Connects one or more non-backbone areas to Area 0. Maintains separate LSDBs per attached area.
- **Autonomous System Boundary Router (ASBR):** Connects the OSPF domain to external routing domains (e.g., BGP, EIGRP, static routes).
- **Internal Router:** All interfaces reside within a single area.
- **Design Rules:**
 1. Area 0 must be present in multi-area topologies.
 2. Non-backbone areas must connect to Area 0 directly (or via virtual link).
 3. Routers within the same area maintain an identical Link-State Database (LSDB).

3. OSPF Packet Types

OSPF operates using five Link-State Packet types:

Type	Packet Name	Function
1	Hello	Discovers neighbors, verifies bidirectional communication, elects DR/BDR.
2	Database Description (DBD)	Summarizes local LSDB contents for database synchronization.
3	Link-State Request (LSR)	Requests specific, full LSAs missing after DBD comparison.
4	Link-State Update (LSU)	Delivers requested LSAs (one LSU carries multiple LSAs).
5	Link-State Ack (LSAck)	Explicitly acknowledges receipt of LSU packets.

4. Router ID (RID) Selection

Every OSPF router requires a unique 32-bit Router ID. Selection priority:

1. Manually configured RID: `router-id <x.x.x.x>`
2. Highest IPv4 address on any active loopback interface.
3. Highest IPv4 address on any active physical interface.

5. Neighbor Adjacency States

Routers progress through seven states to establish full neighbor adjacency:

1. **Down:** No Hello packets received from neighbor.
2. **Init:** Hello received from neighbor, but local RID is not yet in the neighbor's Hello list.
3. **2-Way:** Bidirectional communication established (local RID seen in neighbor's Hello). DR/BDR election happens here. Non-DR routers remain in 2-Way/DR0THER.
4. **ExStart:** Master/Slave relationship and initial sequence numbers negotiated. Higher RID router becomes Master.
5. **Exchange:** Routers exchange DBD packets describing their LSDBs.
6. **Loading:** Routers send LSRs for missing/outdated LSAs and receive LSUs.
7. **Full:** Databases are completely synchronized; normal routing state.

6. Adjacency Requirements (Checklist)

Neighbors will fail to form adjacency if any of the following parameters mismatch:

- **Area ID:** Must be identical on the shared segment.
- **Subnet & Subnet Mask:** Must match exactly on the interface.
- **Hello and Dead Timers:** Default is 10s Hello / 40s Dead on broadcast links.
- **Authentication:** Type (Null, Cleartext, MD5/SHA) and key/password must match.
- **Area Flags:** Area type parameters (e.g., Stub, NSSA) must agree.
- **Interface MTU:** MTU mismatch causes routers to hang in ExStart or Exchange.
- **Router ID:** Must be globally unique across the entire OSPF domain.

7. Core OSPF Tables

Table	Cisco Command	Purpose
Neighbor Table	<code>show ip ospf neighbor</code>	Directly connected neighbors and adjacency states.
Database Table (LSDB)	<code>show ip ospf database</code>	Complete topological roadmap of links and costs.
Routing Table	<code>show ip route ospf</code>	Best paths installed by Dijkstra algorithm for forwarding.

8. LSA Types Reference

Type	Name	Origin	Description
1	Router LSA	Every router	Describes local links and interface costs within the area.
2	Network LSA	DR	Describes multi-access network segment and attached routers.
3	Network Summary	ABR	Advertises inter-area routes into other areas (0 IA).
4	ASBR Summary	ABR	Advertises reachability to an ASBR across areas.
5	AS External	ASBR	Advertises routes redistributed from outside the OSPF domain (0 E1 / 0 E2).
7	NSSA External	ASBR in NSSA	Carries external routes inside NSSA; translated to Type 5 by ABR.

9. Cisco IOS Configuration & Verification

Configuration Syntax

```
! Method 1: Global process network statement
Router(config)# router ospf 1
Router(config-router)# router-id 1.1.1.1
Router(config-router)# network 10.1.2.0 0.0.0.255 area 0
```

```
! Method 2: Interface-level activation (recommended)
```

```
Router(config)# interface GigabitEthernet0/0  
Router(config-if)# ip ospf 1 area 0
```

Key Verification Commands

- `show ip protocols` : Routing protocol parameters, process ID, timers, networks.
- `show ip ospf neighbor` : Neighbor status, adjacency state (2-Way/Full), DR/BDR roles.
- `show ip ospf interface brief` : Enabled interfaces, assigned areas, link costs, network types.
- `show ip ospf database` : LSDB contents and LSA type details.
- `show ip route ospf` : Forwarding table entries learned via OSPF.

10. Acronyms & Abbreviations Reference

- **OSPF**: Open Shortest Path First | **SPF**: Shortest Path First | **LSA**: Link-State Advertisement
- **LSDB**: Link-State Database | **LSP**: Link-State Packet | **DBD**: Database Description
- **LSR**: Link-State Request | **LSU**: Link-State Update | **LSAck**: Link-State Acknowledgment
- **RID**: Router ID | **DR**: Designated Router | **BDR**: Backup Designated Router
- **ABR**: Area Border Router | **ASBR**: Autonomous System Boundary Router | **NSSA**: Not-So-Stubby Area
- **MTU**: Maximum Transmission Unit | **ECMP**: Equal-Cost Multi-Path | **AD**: Administrative Distance